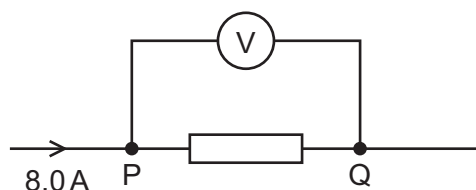


26 Which expression is used to calculate electrical resistance?

- A charge $\div$ potential difference
- B current $\div$ potential difference
- C potential difference $\div$ charge
- D potential difference $\div$ current

27 The diagram shows part of an electric circuit. The reading on the voltmeter is 16 V. The current in the resistor is 8.0 A.



One coulomb of charge flows from P to Q through the resistor.

How much energy is transferred in the resistor?

- A 2.0 J
- B 8.0 J
- C 16 J
- D 128 J

28 What is the combined resistance of a $3.0\ \Omega$ resistor and a $6.0\ \Omega$ resistor connected in parallel?

- A $2.0\ \Omega$
- B $4.5\ \Omega$
- C $9.0\ \Omega$
- D $18\ \Omega$

29 A $5.0\ \Omega$ resistor is connected in series with a $10\ \Omega$ resistor.

There is a constant current in the $5.0\ \Omega$ resistor.

Which statement about the two resistors is correct?

- A The current is greater in the $5.0\ \Omega$ resistor.
- B The current is greater in the $10\ \Omega$ resistor.
- C The potential difference (p.d.) is greater across the $5.0\ \Omega$ resistor.
- D The potential difference (p.d.) is greater across the $10\ \Omega$ resistor.

30 Which statement explains where the switch for an electric lamp is connected?

- A The switch is in **either** the live wire **or** in the neutral wire as both wires carry the current.
- B The switch is in the earth wire to prevent the fuse from blowing when the lamp is in use.
- C The switch is in the live wire so that the lamp is disconnected from the mains supply when the switch is off.
- D The switch is in the neutral wire to prevent current leakage when the lamp is switched off.